

LGDF-Net: Local and Global Feature-Based Dual-Branch Fusion Networks for Deepfake Detection

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Abstract— With the rapid development of Deepfake technology, social security is facing great challenges. Although numerous Deepfake detection algorithms based on traditional CNN frameworks perform well on specific datasets, they still suffer from overfitting due to an over-reliance on localized artifact information. This limitation leads to degraded detection performance across diverse datasets. To address this issue, this study proposes a dual-branch fusion network called LGDF-Net. LGDF-Net uses a dual-branch structure to process the local artifact features and global texture features generated by Deepfake separately, preserving their unique characteristics. Specifically, the local compression branch utilizes a specially designed local compression module (LCM) that allows the network to focus more accurately on key regions of localized artifacts in Deepfake faces. The global expansion branch enhances the analysis of the global facial context through a global expansion module (GEM), which captures image context information and subtle texture features more comprehensively. Additionally, the proposed multi-scale feature extraction module (MSFE) delves into image features at various scales, enriching the extraction of detailed information. Finally, the multi-level feature fusion strategy (MLFF) improves the integration of local and global features through multiple layers, enabling the network to learn the intrinsic connections between these two types of features. A series of experimental validations demonstrate that the proposed scheme outperforms many existing detection networks in terms of accuracy and generalization ability.

Index Terms— Deepfake detection, dual-branching framework, multi-scale feature extraction, multi-level feature fusion.

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I. INTRODUCTION

WITH the rapid development of deep learning in image synthesis, it has brought significant convenience to various fields. However, its misuse has posed considerable challenges to society, particularly with the rise of Deepfake. Advances in GANs [1] and diffusion models [2] have enabled increasingly sophisticated Deepfake forgeries, including the manipulation of facial features, expressions, and even the complete exchange of faces. These fabricated images not only are realistic but also produce subtle artifacts that are challenging to detect, and it significantly diminishes the effectiveness of traditional detection techniques. Furthermore, such deceptive information poses threats to the society when it is exploited for extortion and political unrest, which leads to widespread security concerns. Consequently, there is an urgent need to develop Deepfake detection techniques that exhibit high accuracy and robust generalization capabilities to effectively mitigate these potential risks.

Deepfake techniques can be classified into two main categories: FaceSwap and facial attribute forgery manipulation. The forgery traces generated by Deepfake are not only restricted to specific facial regions but may also be dispersed across the entire face. As shown in Fig. 1, some forgery traces are primarily concentrated in key facial areas, such as the eyes and mouth, while other forgery traces are distributed globally across the face. The artifact traces in key facial regions can be regarded as local artifact information, while the subtle unnatural texture changes in the global area are considered as global texture information. These artifact traces and texture changes are prevalent in images of fake faces. However, many existing Deepfake detection methods rely heavily on local artifacts as the primary detection cues and are typically built on mainstream backbone networks designed for general feature extraction [3], [4], [5]. Although these methods achieve some success in detection, the backbone networks are not specifically designed for Deepfake detection. They focus more on extracting image content and neglect the subtle forgery traces introduced by Deepfakes, leading to poor generalization when confronted with more advanced forgery techniques. Inspired by [6], [7], and [8], as Deepfake technology continues to evolve, the effectiveness of the existing detection methods will be constrained by these backbone networks. Therefore, there is an urgent need for the development of network

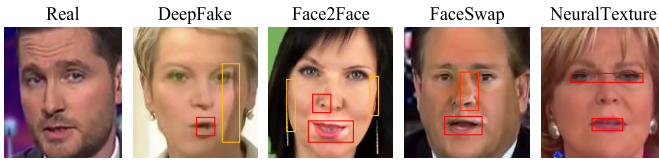


Fig. 1. Schematic representation of the forgery traces, yellow boxes marking the global texture changes produced during the forgery process and red boxes used to mark the local artifacts produced by the forgery.

architectures specifically designed for Deepfake detection to more efficiently extract forged facial features.

Current mainstream research primarily focuses on enhancing the extraction of localized artifact information [9], [10], [11], [12]. While local artifacts are important cues for Deepfake detection, over-reliance on them can impair the generalization ability of the network and ignore other potential forgery cues in the face. As a result, many studies have begun to explore Deepfake detection from a new perspective to reduce dependence on local artifacts. For instance, literature [13] enhanced generalization by developing the integrated face analysis network (iFAN), which learns correlations across multiple facial analysis tasks. Meanwhile, network performance is improved by minimizing interference from irrelevant factors, such as background or identity information [14], [15], [16]. Additionally, some studies [17], [18], [19] expanded the scope of forgery cue extraction by exploring global texture information from often-neglected facial regions, providing a more comprehensive approach to identify forgery traces. Although the significance of both local artifact and global texture is recognized as cues in Deepfake detection [20], [21], there still have limitations. For instance, a single-branch network is utilized to simultaneously extract two types of features with distinct characteristics in [20], which limits the effectiveness of feature extraction.

Local artifact information is a critical visual feature in Deepfake detection, as it effectively captures detailed distortions, while global texture information helps evaluate the consistency and naturalness of the image. Compared to the potential limitations of single-branch architectures in handling complex forged images, the dual-branch architecture extracts local and global features separately through independent branches. This enables a deeper understanding of different types of features while both local and global information can be flexibly fused. As a result, the dual-branch architecture can effectively mitigate the feature confusion problem and enhance the network's detection performance.

Based on the above motivation, LGDF-Net is proposed for Deepfake detection. To address the distinct characteristics of local artifacts and global textures, specialized modules—Local Compression Module (LCM) and Global Expansion Module (GEM)—are designed, forming the local compression branch and global expansion branch within the network framework. This design enhances the network's sensitivity to both types of forgery cues. Additionally, a multi-level feature fusion (MLFF) strategy is introduced to overcome the limitations of local artifact extraction and the insufficient focus on key information during global feature extraction. By fusing local artifact features into global texture features across different

network layers, a powerful joint feature representation is created. To further improve detection efficiency and accuracy, a Multi-Scale Feature Extraction (MSFE) module is developed to enrich feature extraction and mitigate the loss of critical information.

In summary, the main contributions of this study include:

- A dual-branch framework LGDF-Net comprising LCM and GEM modules is designed. The dual-branch structure processes local artifactual cues and global texture information separately, facilitating more accurate learning of the unique features associated with each type. It can achieve accurate detection of deep artifacts with robust cross-dataset generalization.
- A multi-scale feature extraction module (MSFE) is created to integrate diverse feature information from images. By employing a multi-branch structure with various types of feature extraction layers, the module can extract richer detailed features and enhance the model's representational capability.
- A multi-level feature fusion strategy (MLFF) is developed to strengthen the connection between these two types of features. The combination of local artifact features and global texture features are realized in multiple hierarchical scale, so that the network can acquire the appropriate feature representation.

II. RELATED WORK

As Deepfake technology uses complex generative models to create highly realistic fake images and videos, the gap between the generated images and real facial images is narrowing. The rapid advancement of this technology poses new challenges for information security and the detection of fake information. Currently, Deepfake detection techniques have achieved significant progress, and they can be classified into the following three categories based on different detection clues.

A. Detection Based on Local Artifacts

It focuses on facial details and distortions in local features, effectively capturing subtle traces of forgery. Early studies performed Deepfake detection by manually extracting visual feature differences in images, such as inconsistencies in head pose [22] or eye blink anomalies [23]. With the continuous advancement of Deepfake technology, these traditional methods confront increasing challenges. Deepfake detection is generally considered as a binary classification problem and some advanced image classification models have been utilized for detection, such as VGGNet [3] and XceptionNet [4]. However, these backbone networks inherited from the image classification task are more concerned with semantic feature extraction within the image and fail to adequately consider the critical local subtle artifacts in Deepfake detection task. Therefore, several studies [24], [25], [26], [27] have focused on local artifacts prevalent in Deepfake by enhancing network structure. For example, Zhao et al. [28] used a multi-attention mechanism to enhance the focus on local texture and high-level semantic features. Guo et al. [14] utilized convolution to predict manipulation traces in images and backpropagation to update the weights, thereby expanding

artifact traces. Tan et al. [29] used a pre-trained CNN model to convert an image into gradients, and then they are used to render generalized artifacts. Although these methods have improved detection accuracy to some extent, the visual feature differences in generated forgery images are diminishing with the evolution of Deepfake technology, and the limitations of an over-reliance on local artifactual cues are emerged.

B. Detection Based on Specific Clues

It analyzes the frequency domain, color space and noise characteristics of forged images to identify distinctive indicators. Since forged images generated by Deepfake technique may leave some clues on these aspects, it is possible to determine whether an image is forged or not. For example, Liu et al. [15] combined spatial images and phase spectra to capture the apparent changes generated during the forgery process. Li et al. [30] built a frequency-aware framework that enhances interclass differences in the embedding space to capture the frequency of Deepfake changes from different inputs. Residual signals from chromaticity components of multi-color spaces are used as network inputs in [31] to extract forged common features. In [32], a wavelet scattering network is used to extract the YCbCr signal from the image color space, which is converted into a comparison score to predict the final result by processing the edge discontinuities generated by the distortions in each color channel. Luo et al. [33] removed the color texture by image noise to reveal the difference between the real and manipulated regions. Zhou et al. [34] designed a two-stream network to capture face manipulation artifacts and local noise residues. Shiohara and Yamasaki [35] independently developed a hybrid image dataset by adopting the synthetic approach in [36], which effectively avoids the problem of overfitting to a single artifact. These specific cues enhance the detection capabilities of the network and reduce reliance on local artifacts. However, they can be corrupted during image compression, which adversely affecting performance.

C. Detection Based on Global Information

It assesses the consistency and naturalness of the overall image by analyzing global features to determine the authenticity of the face. Although local artifacts are widely regarded as key detection cues, recent research indicates that changes in global information should also be considered. Global information detection focuses on analyzing overall facial texture changes and the relationships between localities in image or video. For example, in [37] and [38], the relationships between localities are analyzed globally by using Transformer. After that, these local features are no longer viewed independently as in previous studies. Chen et al. [39] learned local relationships by measuring the similarity between features in local regions, facilitating a more comprehensive global analysis of forged generalized features from a novel perspective. Shau et al. [40] expanded the Deepfake detection range of the network by a two-stream network instead of limiting it to a few dominant detection regions, thereby capturing a broader array of Deepfake detection cues. Yang et al. [17]

recognized the potential of global texture information to distinguish texture of forged faces from multiple scales using centroid difference convolution (CDC) and spatial pyramid pooling differences. Liu et al. [19] utilized global image texture features for Deepfake detection to achieve extremely high accuracy. Meanwhile, studies in [20], [41], and [42] explored the joint role of global texture information and local artifact information to improve network performance. In [20], a multi-classification task-based detection framework was proposed to expose subtle texture changes induced by Deepfakes on a global scale and to differentiate between types of Deepfakes. Meanwhile, Guo et al. [41] conducted global texture mining while capturing local artifact features to reveal global subtle texture changes caused by Deepfakes. Although the above studies have considered the importance of local artifact information and global texture information in Deepfake detection, the distinctive characteristics of these two types of detection signals are not fully exploited. These studies typically rely on the information extraction modules from existing image classification networks without incorporating a dedicated design for detecting Deepfake specific forgery cues. Furthermore, in terms of fusion between local artifacts and global texture features, most of these studies usually integrate them at the final stage of the network using simple operations, such as concatenation or summation, which limits the full potential of these features.

In this study, the extraction of local artifacts and global texture features are optimized through a dual-branch architecture, allowing the network to comprehensively identify unnatural facial regions. To preserve feature information into these two branches as much as possible, MSFE module is designed to enhance detail extraction and minimize information loss. Subsequently, through MLFF, it effectively mines and learns robust representations of these feature type. It not only addresses the limitations of prior studies but also offers a more effective solution for Deepfake detection, promoting further advancements in this field.

III. PROPOSED SCHEME

The overview of LGDF-Net is shown in Fig.2. It utilizes a dual-branch architecture combined with MSFE module and MLFF strategy to enhance the network's accuracy and generalization ability. Targeted extraction of local artifact and global texture features is performed through local compression and global expansion branches, preserving their unique characteristics. Additionally, the MSFE module extracts image features from multiple perspectives, providing rich details for subsequent network learning. The MLFF strategy aggregates local artifact and global texture information at different levels, adjusts the network's attention to these feature types and continuously optimizes their representation. The multi-scale feature extraction module, local compression branch, global expansion branch, and multi-level feature fusion strategy are elaborated in the following.

A. Multi-Scale Feature Extraction Module

To compensate for the information loss caused by traditional CNNs during feature extraction and to enrich the diversity of

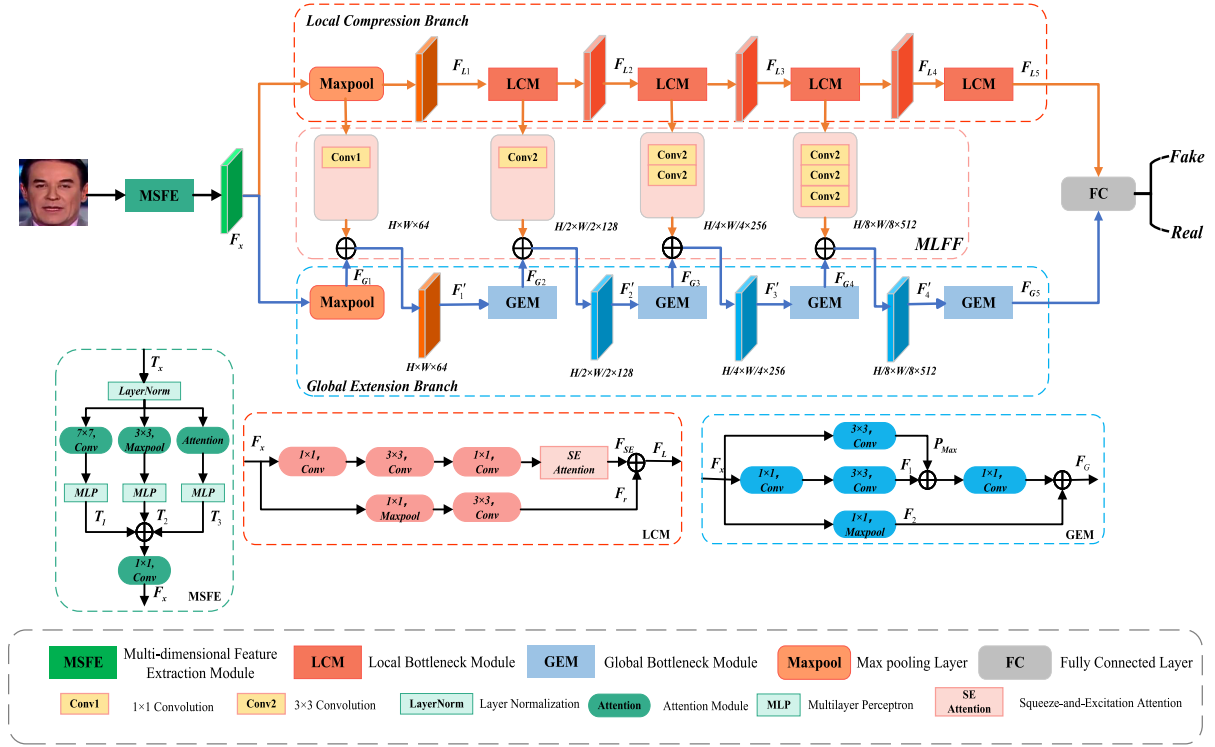


Fig. 2. The general framework of LGDF-Net. This dual-stream network consists of three main modules, MSFE, LCM, and GEM. The details of each module are shown in Fig.

learned feature types, a multi-scale feature extraction (MSFE) module is designed. This module consists of a convolutional branch, a max pooling branch, and an attention branch, and the attention branch is based on the self-attention mechanism [43]. Each branch provides distinct spatial and semantic representations. By aggregating these varied representations, the detail loss associated with standard convolution can be minimized. This enables the network to learn finer details and richer facial profile information, significantly enhancing the clarity and resolution of the feature maps extracted by the network.

The MSFE module consists of three main parallel branches, and each of which contains two key components. The first of which is a feature extraction layer for different image information, consisting of a 7×7 convolutional with stride of 2 and padding of 3, a 3×3 max pooling with stride of 2 and padding of 1, and a self-attention module, respectively. The second part of the multi-layer perceptron (MLP) is responsible for processing different types of features extracted from each feature extraction layer separately. As shown in Fig.2, for the input image, it first extracts the preliminary feature map $T(x) \in R^{C \times H \times W}$ by 7×7 convolutional layer and then input it to the MSFE module for depth processing. In the MSFE module, the feature maps $T(x)$ are first normalized by a normalization layer, which are randomly initialized as learnable parameters. These parameters are updated during training to adapt to the intrinsic distribution of the data, providing a canonical and stable input benchmark for subsequent operations. Subsequently, these processed features are fed into the three branches corresponding to the convolutional, max pooling, and attention layers, and the pixel-to-pixel interactions are performed in their respective MLP to obtain the

higher-level abstract feature mappings $T_i(x)$, $i \in \{1, 2, 3\}$. Here, $T_1(x)$, $T_2(x)$, and $T_3(x)$ represent the output features of the convolutional, max pooling, and the attention branches, with channel numbers of 64, 32, and 32, respectively. $\sigma(\cdot)$ denotes the multi-layer perceptron. $F_C(x)$, $F_M(x)$, $F_A(x)$ represent the convolutional layer, the max pooling layer and the attention layer, respectively. The operation flow of each branch is shown by

$$T_i(x) = \sigma(F_j(T(x))), i = 1, 2, 3, j = C, M, A \quad (1)$$

After processing and optimization in each branch, the feature mappings T_1 , T_2 , T_3 are up-sampled to achieve uniformity in dimension. Subsequently, these features are spliced in the channel dimension to achieve effective fusion of the image detail information. Finally, the output feature is completed by a convolutional layer with a kernel of 1×1 , stride of 1, and padding of 0, which is represented by

$$F(x) = \text{Conv}_{1 \times 1}(T_1 + T_2 + T_3), \quad (2)$$

where $F(x)$ denotes the output of the MSFE module, $\text{Conv}_i(\cdot)$ denotes the convolution operation. The subscript i denotes the size of the corresponding convolutional kernel. The optimized feature $F(x)$ output by the multi-dimensional feature extraction module will provide key support for the subsequent dual-branch Deepfake detection task to ensure the accuracy of the detection.

Fig.3 shows the visualization of the image features extracted by each branch in the MSFE module. It can be found that the features extracted by different branches present significant differences. Specifically, the convolutional branch focuses on the extraction of shallow features, so the extracted shallow

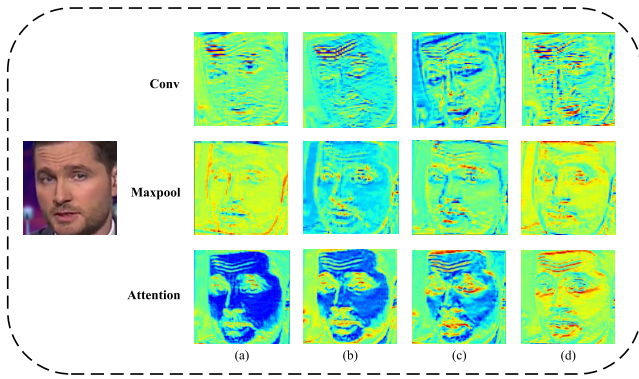


Fig. 3. Visualized feature maps extracted using each branch of MSFE, (a) ~ (d) correspond to the output of feature maps on different channels.

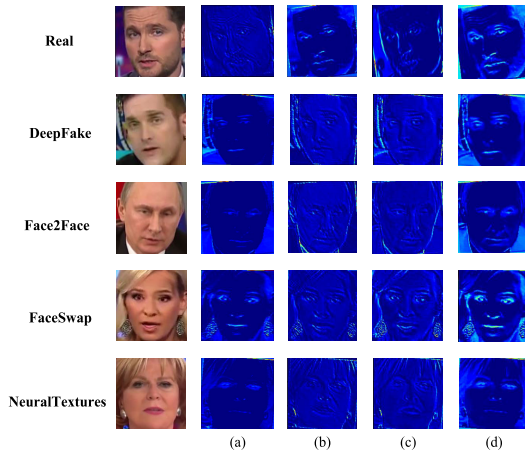


Fig. 4. Visualized feature maps extracted using standard convolution, (a) ~ (d) correspond to the output of the feature maps on different channels.

information of the image is richer, as shown in Fig.3 (b)~(d). The maximum pooling branch focuses on capturing significant artifacts in the image, so the eyes, mouth, and facial contour parts of the face are more sensitive, as shown in Fig.3 (a)~(c). The attention branch is dedicated to capturing key feature information from each local region of the face, and it pays more attention to the key features in each local region of the face, instead of limiting itself to the five facial features, so the deficiencies of the max-pooling branch is enhanced, as shown in Fig.3 (c)~(d).

Additionally, Fig.4 and Fig.5 display the visualized feature maps extracted by standard convolution layers and MSFE modules, respectively. It can be found that the feature information extracted by standard convolution is relatively limited, which indicates a restricted ability to capture important facial details and accurately identify local key information and facial contours, as shown in Fig.4(a)~(d). In contrast, the feature maps extracted by the MSFE module exhibit richer and more detailed facial features, as illustrated in Fig.5(a)~(d). This improvement is attributed to the MSFE module's capability to enhance image information extraction and reduce the interference of irrelevant information by aggregating multiple types of features through its multiple branches.

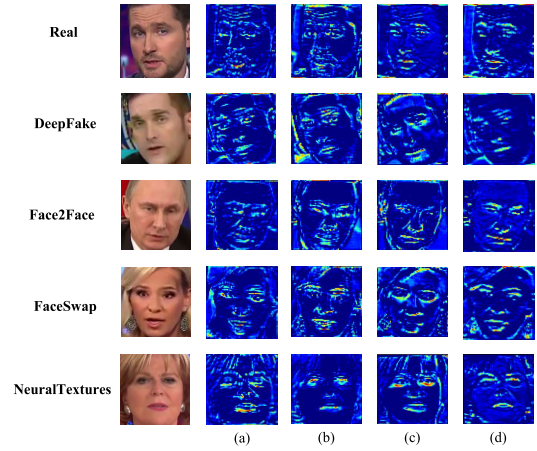


Fig. 5. Visualized feature maps extracted using MSFE, (a) ~ (d) correspond to the output of feature maps on different channels.

B. Local Compression Branch

In the Deepfake detection task, local artifact features serve as crucial detection cues. To enhance the network's ability to model local artifact features in the forged image, the network's attention is concentrated on key local artifact regions to minimize the interference of irrelevant information. To achieve this, a local compression module (LCM) that combines the residual structure [5] with the bottleneck structure [44] is designed. Local compression branch is formed by cascading LCM modules. In the LCM, the number of channels in the convolution is modified through dimensionality reduction followed by dimensionality enhancement, which helps to control the network's extraction space in regions with a high concentration of forged artifacts. It reduces the risk of propagating irrelevant redundant information within the network, compresses the sensory field without introducing additional parameters and avoids the computational complexity associated with excessive network depth.

The LCM module consists of an inverse bottleneck structure, a residual structure, and a SE attention module [45], as shown in Fig.2. The inverse bottleneck structure consists of three convolutional layers using 1×1 , 3×3 , and 1×1 convolutional kernels, all with stride of 1, and padding 0, 1, 0, respectively. The residual structure consists of a max pooling layer and a convolutional layer using 1×1 and 3×3 convolutional kernels, all with stride of 1, and padding of 0, 1, respectively. The SE attention module is responsible for the weighting of the feature channels with respect to their importance in order to highlight important features. Taking $F(x)$ as the input to the LCM module, the compression and expansion of the input feature channel is first realized by a sequence of convolutional layers with an inverse bottleneck structure. Then, the SE attention module adjusts the weights of the compressed and expanded features to get the final output, and the whole process can be expressed as

$$F_{SE} = SE(Conv_{1 \times 1}(Conv_{3 \times 3}(Conv_{1 \times 1}(F(x))))), \quad (3)$$

where $F_{SE}(x)$ denotes the output features processed by the SE attention mechanism, $SE(\cdot)$ represents the SE attention

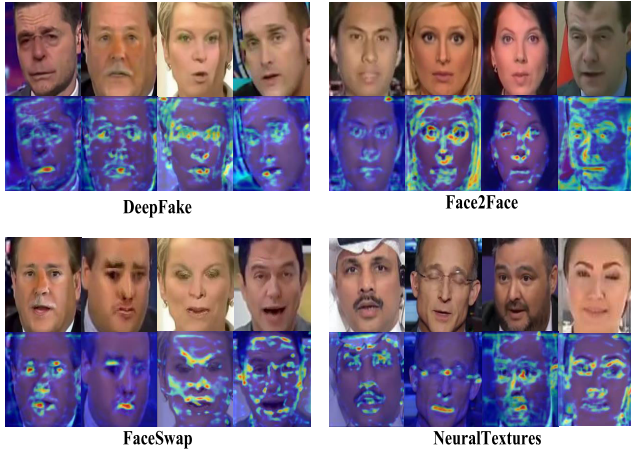


Fig. 6. Grad-CAM mapping diagrams for the LCM module under four forgery types in FaceForensics++.

mechanism, and $\text{Conv}_i(\cdot)$ denotes the convolution operation. The subscript i denotes the size of the corresponding convolutional kernel. To compensate for the loss of information, the residual output $F_r(x)$ is obtained by residual structuring the original input features $F(x)$ through 3×3 convolution and 1×1 max pooling, which is denoted by

$$F_r(x) = \text{Conv}_{3 \times 3}(\text{P}_{\text{Max}}(F(x))), \quad (4)$$

where $\text{P}_{\text{Max}}(\cdot)$ denotes the max pooling layer. Finally, F_{SE} is combined with F_r obtained via the residual structure to supplement the detail information to obtain the final output of the LCM $F_L(x)$, which can be expressed as

$$F_L(x) = F_{SE}(x) + F_r(x). \quad (5)$$

Throughout the local compression branch, the input features will be processed by a cascade of four LCM modules to refine the capture of local regions of the facial image. The LCM module is able to focus on facial features in different localized regions to ensure that the final output has higher discriminative power, which further improves the detection performance of the entire network.

The Grad-CAM in Fig.6 shows the focus of the LCM module's attention on each region in the face image. It can be seen that the LCM focuses its attention on the localized five-feature parts where artifacts frequently appear in Face2Face and NeuralTextures, while the LCM focuses its attention on the facial distortion traces caused by the artifacts, as well as the irregular contours of the face in Deepfake and FaceSwap. This also verifies that the designed LCM module is effective in forcing the model to capture the local artifactual cues of the face in a more compact feature space by reducing the number of channels, and then further assigning weights to the extracted features to get the key features through the SE module. Therefore, the local compression branch using the LCM module can effectively highlight the key Deepfake cues, enhance the sensitivity of the network to these forged features, and further improve the model performance.

C. Global Expansion Branch

Subtle texture changes are typically randomly distributed across various regions of the face. By expanding the network's search range, these facial texture changes can be captured comprehensively, and the limitations of the local extraction process can be addressed. Furthermore, the feature information obtained from the global search aids the network in accurately targeting camouflage regions, thereby enhancing detection accuracy. To facilitate this, a global expansion module (GEM) is designed based on the residuals and inverted bottleneck structure [44] to constitute the global expansion branch. By increasing the number of channels to enhance the perceptual domain of feature extraction, the network can better understand the overall structure of the face through contextual information, and the ability to capture subtle global texture changes. Additionally, to further enhance the learning capability of the global expansion branch, the features output from the local and global branches of the previous layer are optimally fused as inputs to the GEM module.

The GEM module contains two standard convolutions with kernel of 1×1 and two kernels of 3×3 , which are corresponded to strides 1 and 2 with padding 0 and 1, and max pooling layer with kernel 3×3 , stride 2, and padding 1, as shown in Fig.2. $F(x)$ is taken as the input to the GEM module, where $F(x)$ is first dimensionally scaled by 1×1 and 3×3 convolutional layers. During this dimensionality adjustment process, the number of channels is increased and then decreased to increase the network sensory field for deeper understanding of the contextual information. It can be represented as

$$F_1(x) = \text{Conv}_{3 \times 3}(\text{Conv}_{1 \times 1}(F(x))), \quad (6)$$

where $F_1(x)$ denotes the dimensionally adjusted output feature. $F_1(x)$ is combined with $\text{P}_{\text{Max}}(F(x))$, which is restored to the original number of channels by a 1×1 convolution operation, and the enhancement feature $F_2(x)$ to finally obtain the output $F_G(x)$. Specifically, it can be expressed by

$$F_2(x) = \text{Conv}_{3 \times 3}(F(x)), \quad (7)$$

$$F_G(x) = \text{Conv}_{1 \times 1}(F_1(x) + \text{P}_{\text{Max}}(F(x))) + F_2(x). \quad (8)$$

In the global expansion branch, the input information is optimized by using the GEM module to reduce the width and height of the input feature map, which helps to focus on a wider range of pixel, thus reducing unwanted local noise. This process can be represented by

$$F_{Gi}(x) = \text{GEM}_{i-1}(F'_{i-1}(x)), i = 2, 3, 4, 5. \quad (9)$$

where $F'_i(i = 1, 2, 3, 4)$ denote the optimized features inputted into the GEM module, and $F_{Gj}(j = 2, 3, 4, 5)$ are the output features obtained by the GEM. Their sizes are $128 \times 38 \times 38$, $256 \times 19 \times 19$, $512 \times 10 \times 10$ and $1024 \times 5 \times 5$, respectively. The global expansion branch digs deeper into the high-level texture information in the image features by cascading GEM modules and effectively extracts the global subtle texture changes.

Fig.7 shows the global texture information extracted by the GEM module from authentic faces and various types of counterfeit faces. The GEM can effectively extract fine

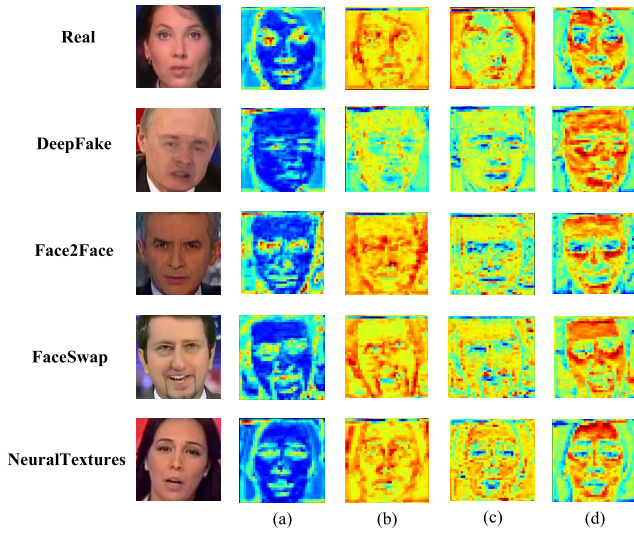


Fig. 7. Global texture extraction of real faces, (a) ~ (d) are visualized feature maps extracted by the global bottleneck module.

facial contours in both real and fake faces, and preserve sensory information while also retain rich texture details from non-critical facial regions, as depicted in Fig. 7(c)~(d). This effect is mainly due to the fact that the GEM expands the number of channels based on the bottleneck structure, and it enhances the network's receptive field and its ability to capture intricate features. Additionally, the incorporation of residual structures in the GEM facilitates information supplementation, thereby deepening contextual learning.

D. Multi-Level Feature Fusion Strategy

In contrast to the traditional approach that simply splices local artifacts and global texture at the end of the network, a multi-level feature fusion strategy (MLFF) that learns the abstract features of these two types of cues at various levels is proposed. This strategy adjusts the network's attention allocation to different feature types, thereby providing a more discriminative feature representation.

As shown in Fig. 2, the MLFF strategy first processes the output F_{L1} from the first layer of the local branch using a 1×1 convolution with a stride of 1 and padding of 0. This output is then combined with the output F_{G1} from the first layer of the global branch to yield F'_1 , which is sent to the second layer of the global branch to produce the output feature F_{G2} . Secondly, the output F_{L2} from the second layer of the local branch is processed with a 3×3 convolutional layer with a stride of 2 and padding of 1, and then its size is reduced to half of the original. Then, it is combined F_{G2} to generate F'_2 , which is fed into the third layer of the global branch to obtain F_{G3} . The output F_{L3} from the third layer of the local branch undergoes processing through two 3×3 convolutional layers with a stride of 2 and padding of 1, so its feature map size is reduced to one-fourth of the original. It is combined with F_{G3} to produce F'_3 , which serves as the input for the fourth layer of the global branch. Finally, the output F_{L4} from the fourth layer of the local branch is processed using three 3×3 convolutional layers with a stride of 2 and a

padding of 1, and its size is reduced to one-eighth of the original. This output is combined with F_{G4} , which is the output from the fourth layer of the global branch, to obtain F'_4 . Finally, it is input into the fifth layer of the global branch for the final output.

The fusion strategy eliminates the impact of unnecessary noise by reducing the size of the feature maps, thereby enhancing the focus on local features. Furthermore, by integrating local artifact features with global texture features, forgery cues in the manipulated face can be extracted more accurately and comprehensively.

IV. EXPERIMENTS

A. Datasets

Here, three mainstream datasets FF++ [46], DFDC [47], and Celeb-DF [48] are chosen for experimental evaluation. Since all three datasets are video datasets, it is necessary to perform preprocessing on them. Firstly, the face locations in the video frames are marked using the face recognition library MTCNN, and then these coordinates corresponding to faces are cropped using the CV2 library. After that, the extracted face images are resized to 299×299 for evaluation. Considering the inconsistency in the number of real and fake videos in the dataset, the number of extracted frames has been adjusted accordingly for different datasets.

FF++ [46] consists of 1,000 real videos and 4,000 fake videos generated by four typical face forgery methods, including Deepfake, Face2Face, FaceSwap, and NeuralTextures. Three versions are included in FF++: the original dataset, the compressed high-quality dataset FF++ (HQ), and the compressed low dataset FF++ (LQ). To better detect the performance of the model, the more challenging FF++ (HQ) and FF++ (LQ) datasets are experimentally chosen.

DFDC [47] is a large-scale fake face video database developed for Deepfake Detection Challenge, and it contains 1131 real videos and 4119 fake videos. Various advanced Deepfake techniques, and the real videos are shot in different real-life scenarios and have partially faked Deepfake faces.

Celeb-DF [48] contains 590 real videos and 5639 fake videos, and it is a currently popular database of high-quality face fakes. By improving the Deepfake method, the visual artifacts presented in the video is reduced. The generated realistic faces make Deepfake detection very challenging, and they are commonly used to evaluate the generalization ability of the model.

B. Experiment Settings

1) *Evaluation Metrics*: To comprehensively evaluate the performance of the methods, accuracy (ACC) and area under the curve (AUC) are used as the main evaluation criteria to evaluate performance. ACC represents the percentage of real versus fake images correctly recognized by the classification network, while the AUC metric is the overall quality of the model prediction results, and it reflects the model discriminative power. With these two metrics, it is able to make an objective evaluation of the method performance on the Deepfake detection task.

TABLE I
DEEPPFAKE DETECTION COMPARISON RESULTS FOR INTRA-DATASET EVALUATION

Methods	DFDC		Celeb-DF		FF++ (HQ)		FF++ (LQ)	
	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC
MesoIncep[24]	93.28	0.984	95.66	0.994	82.63	0.919	70.76	0.795
XceptionNet[4]	99.16	1.000	96.21	0.999	97.72	0.998	96.64	0.996
F ³ Net[49]	98.58	0.998	99.88	1.000	99.14	1.000	96.81	0.994
MAT*[28]	-	-	97.92	0.999	97.60	0.993	88.69	0.904
Recce[50]	99.32	1.000	99.30	1.000	97.04	0.998	96.76	0.996
M2TR*[51]	-	-	-	-	98.23	0.995	92.35	0.942
GocNet[52]	98.95	1.000	99.82	1.000	94.84	0.992	91.25	0.970
SFDG*[53]	-	-	99.22	0.999	98.19	0.995	92.28	0.960
LDF-Net*[41]	93.15	0.972	-	-	96.01	0.989	92.32	0.968
LGDF-Net(our)	99.53	1.000	100.00	1.000	97.64	0.997	96.81	0.996

* Indicates direct use of the original data.

2) *Implementation Details*: Given that the sizes and shapes of actual faces vary in different video datasets, to ensure the consistency of the input data, MTCNN is used to detect faces in the video dataset, and the faces used in the experiments are cropped to 299×299 . In addition, several commonly used data enhancement techniques are introduced to improve the generalization ability of the model. In the training phase, SGD optimizer with default parameters and an initial learning rate of 0.001 is chosen. The learning rate is set to decay to 50 of the original every 5 epochs. All experiments were implemented in the PyTorch framework through NVIDIA RTX 3090 24GB.

C. Experimental Results

1) *Intra-Dataset Performance*: To achieve an objective and fair comparison, MesoIncep [24], XceptionNet [4], F³Net [49], MAT [28], Recce [50], M2TR [51], GocNet [52], SFDG [53], LDF-Net [41] are chosen for comparison, and the experimental results are kept under the same experimental conditions for the selected methods trained from scratch on the same dataset and validated using the same test set. The results are listed in TABLE I.

It can be found that the methods detection accuracy in the FF++ dataset is degraded compared to DFDC and Celeb-DF. This is attributed to the richer face samples and the variety of forgeries available in FF++. However, it can be observed that the proposed scheme achieves high accuracy in Celeb-DF and DFDC, and it indicates that the proposed scheme can adequately learn the spatial features of the forged images. In both versions of the FF++ dataset, LGDF-Net still shows excellent performance, especially in the challenging (LQ) version, and it achieves the optimal results in terms of both ACC and AUC compared to other methods. Despite the fact that F³Net reaches the highest ACC on FF++ (HQ), our proposed framework still achieves good results. Meanwhile, LGDF-Net exceeds it on FF++ (LQ), and this is because the over-compression destroys the frequency cues utilized by F³Net. In contrast, LGDF-Net fully learns the local artifacts and global texture information in the forged face, which complement each other and preserve the forged cues after compression. Thus, the generalization and anti-interference ability of the method are improved.

TABLE II
DEEPPFAKE DETECTION COMPARISON RESULTS
FOR CROSS-DATASET EVALUATION

Methods	Train	Test (Celeb-DF)	
		ACC	AUC
MesoIncep[24]	FF++	64.59	0.688
XceptionNet[4]	FF++	58.95	0.630
F ³ Net[49]	FF++	65.47	0.681
Recce[50]	FF++	66.11	0.680
MAT*[28]	FF++	-	0.674
M2TR*[51]	FF++	-	0.682
GocNet[52]	FF++	66.08	0.682
LDF-Net*[41]	FF++	-	0.657
MSF-Net*[42]	FF++	-	0.674
LGDF-Net(our)	FF++	67.14	0.696

* Indicates direct use of the original data.

From the experimental results, it demonstrates the feasibility of LGDF-Net in coping with different forgeries, and significant performance improvement is achieved. This improvement is mainly attributed to the deep analysis and precise refinement of two key cues, local artifacts features and global texture information, by using the dual-branch structure. In this way, it ensures that the unique properties of these cues are effectively maintained and exploited.

2) *Cross-Dataset Performance*: To evaluate the applicability of LGDF-Net in real-world scenarios, cross-dataset testing is performed to assess its generalization capability on unseen forged data. Specifically, the network is trained on the FF++ (HQ) dataset and then the trained network is tested on the Celeb-DF dataset, the results are listed in TABLE II. Additionally, a series of cross-dataset tests are conducted on the four FF++ (LQ) subsets to further assess its generalization ability across various generation techniques. The corresponding results are presented in TABLE III.

As seen from TABLE II, LGDF-Net demonstrates strong generalization performance on the Celeb-DF dataset, the

TABLE III
CROSS-DATASET EVALUATION RESULTS ON THE FF++ (LQ) DATASET

Training Set	Methods	Testing Set(AUC)			
		DF	F2F	FS	NT
DF	F ³ Net[49]	1.000	0.788	0.823	0.749
	GocNet[52]	1.000	0.723	0.709	0.725
	LGDF-Net(our)	1.000	0.784	0.851	0.759
F2F	F ³ Net[49]	0.865	1.000	0.802	0.828
	GocNet[52]	0.757	0.999	0.682	0.764
	LGDF-Net(our)	0.839	1.000	0.825	0.847
FS	F ³ Net[49]	0.932	0.770	1.000	0.700
	GocNet[52]	0.850	0.720	1.000	0.653
	LGDF-Net(our)	0.926	0.756	1.000	0.671
NT	F ³ Net[49]	0.939	0.924	0.812	0.999
	GocNet[52]	0.897	0.874	0.758	0.994
	LGDF-Net(our)	0.929	0.947	0.829	0.999

accuracy reaches 67.14%, which exceeds Recce with 1.03%. In addition, LGDF-Net also excels in AUC. This mainly benefits from the dual-branch architecture of the LGDF-Net framework, which is specifically designed to effectively extract both local artifact features and global texture features. Furthermore, the MLFF strategy enhances the feature representation through multi-level feature fusion, and optimizes the network's attention to different feature cues, thereby significantly improving the generalization capability for Deepfake detection. TABLE III further demonstrates the excellent generalization ability of LGDF-Net. Its performance in the three subsets of FF++ (LQ) surpasses that of the compared methods. It indicates that LGDF-Net can effectively enhance generalization by adequately extracting different forgery cues from the face separately.

D. Ablation Study

LGDF-Net consists of four components: the MSFE module, the local compression branch composed of LCM, the global expansion branch composed of GEM, and MLFF. To verify the efficacy of these components, a series of ablation experiments are implemented on the DFDC, FF++ (HQ), and F++(LQ) datasets.

TABLE IV presents the results of the quantitative experiments under different combinations of configurations. When the network uses standard convolution rather than the MSFE module to extract features, the detection performance of LGDF-Net significantly degrades despite maintaining both the local compression branch and global expansion branch. It indicates that standard convolution loses a large amount of detail information during feature extraction, which limited the network's ability to learn in subsequent stages and reduces its expressiveness. In contrast, the MSFE module is designed to maximize the retention of key image features from multiple scale, which effectively reduces the information

loss by integrating different types of image information in the space and refines richer detailed features for network learning. In addition, when the network consists of only a local compression branch or global expansion branch, it also can achieve quite excellent detection performance. It proves that the local compression branch based on LCM can accurately highlight key artifact regions and effectively suppress the interference of background noise. The GEM assists global expansion branch to understand the image contextual information, and enhance the extraction of subtle texture information. Moreover, the detection efficacy of the network is significantly enhanced when these two branches act in concert. It verifies that both local artifact features and global texture information are key cues for identifying forgeries. To enhance the network to learn the intrinsic connection between these two types of features, the MLFF strategy is adopted to integrate the local and global features and realize the joint expression of local and global branches through hierarchical fusion. As listed in TABLE IV, it can be seen that the detection network improves the detection accuracy by 1.00% and 0.43% in the FF++(HQ) and F++(LQ) datasets with the help of the fusion strategy, respectively. The above analysis demonstrates the effectiveness of each component of LGDF-Net.

E. Experimental Analysis

1) *Visualization Analysis*: Here, the effectiveness of LGDF-Net is validated through visualization. Grad-CAM [54] and line plots are employed to illustrate the feature extraction capabilities of LGDF-Net and the roles of different configurations in the detection process.

To better understand the impact of local and global branches on detection performance, the accuracy (ACC) of each branch across different training epochs is presented in Fig.8. It illustrates that the overall performance of LGDF-Net surpasses that of any single branch during the training phase. The training accuracy of LGDF-Net continues to improve as the performance of each branch stabilizes with the increase of training epochs. It indicates that the proposed LGDF-Net effectively leverages the advantages of both local and global branches, and superior performance can be achieved by integrating feature information from both, which eventually validates the efficacy of the dual-branch fusion network. Furthermore, the final performance of the global branch exceeds that of the local branch, and it demonstrates that global information extraction has a more significant impact on Deepfake detection.

The Grad-CAM visualization results of different methods on the FF++ dataset are shown in Fig.9. It reveals that most heat maps from other methods focus on regions outside the face and background information, while only a few targets the key areas of the face. In contrast, the heat maps produced by the proposed LGDF-Net for Deepfake (DF) and FaceSwap (FS) effectively emphasize the unnatural regions of the face, and the local artifacts and texture variations are highlighted. Moreover, in NeuralTextures (NT), the heat maps exhibit good coverage of critical facial locations. These findings further demonstrate the superiority of the proposed LGDF-Net.

2) *Computational Efficiency Analysis*: In order to verify the effectiveness of our proposed method in practical

TABLE IV
ABLATION STUDY OF DIFFERENT COMPONENTS ON LGDF-NET

Network	MSFE	LCB	GEB	MLFF	DFDC		FF++ (HQ)		FF++ (LQ)	
					ACC	AUC	ACC	AUC	ACC	AUC
LGDF-Net	✓	✓	✓		98.63	0.999	96.64	0.995	96.38	0.988
		✓	✓	✓	96.85	0.997	96.65	0.995	94.89	0.989
	✓		✓		98.79	0.999	95.13	0.993	93.97	0.989
	✓	✓			97.79	0.999	94.85	0.992	93.47	0.986
	✓	✓	✓	✓	99.53	1.000	97.64	0.997	96.81	0.996

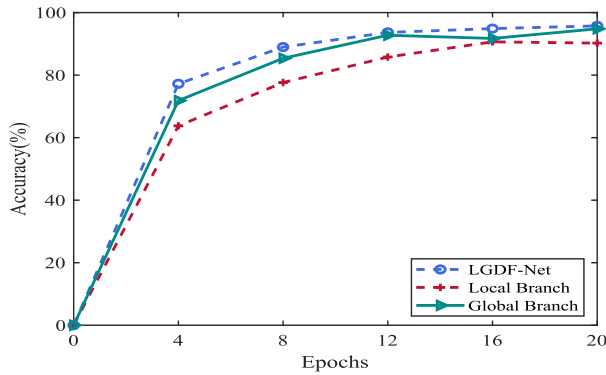


Fig. 8. Visualization of the training accuracy of the proposed scheme under different network configurations.

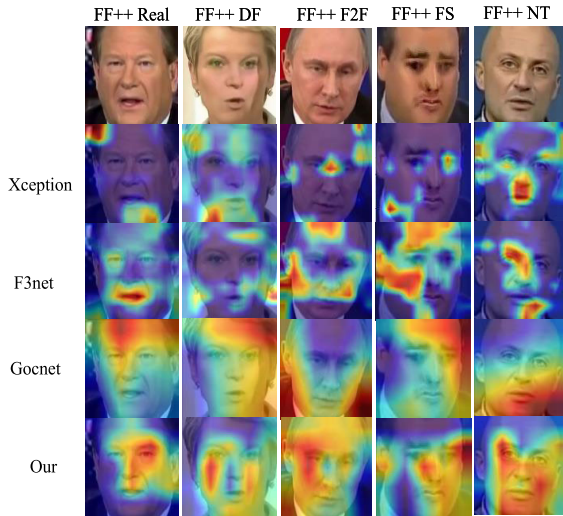


Fig. 9. Grad-CAM visualization of different methods on FF++ dataset.

applications, the computational efficiency of LGDF-Net is compared with some state-of-the-art methods, such as XceptionNet [4], F³Net [49] and Recce [50], in terms of params and inference time. The computational efficiency results are shown in TABLE V.

From TABLE V, it can be seen that the number of parameters in LGDF-Net is larger than those of other methods except F³Net. The increase of parameters is due to the addition of multiple convolutional layers, which were incorporated to enhance the network's feature extraction capability, particularly to facilitate the sufficient extraction of both local and global features at each layer. In terms of inference time, the

TABLE V
COMPARISON OF COMPUTATIONAL EFFICIENCY OF MODELS

Methods	Params	Inference time
XceptionNet[4]	20.81M	9.72ms
F ³ Net[49]	47.96M	16.45ms
Recce[50]	23.78M	80.27ms
LGDF-Net(our)	40.60M	14.09ms

average inference time of a single image on the GPU is calculated. Xception [4] shows the shortest inference time in TABLE V, because Xception [4] is a backbone classification network that does not include additional modules for Deepfake detection. In contrast, LGDF-Net demonstrates a clear advantage over other Deepfake detection methods in terms of inference time. It achieves faster inference speed while maintaining high detection accuracy and generalization ability.

The experimental results demonstrate that the proposed LGDF-Net outperforms existing methods in terms of accuracy and generalization for deepfake detection. Furthermore, its fast inference speed offers significant advantages for real-time detection and large-scale deployment scenarios.

V. CONCLUSION

In this paper, a local and global feature based dual-branch fusion network LGDF-Net is proposed. The network learns the unique properties of local artifact features and global texture features separately, which can compressively capture these two types of Deepfake cues. Among them, the local compression branch composed of LCM modules focuses on key local regions to mine local artifact clues. The global expansion branch composed of GEM modules deepens the ability to understand the image globally through the extracted contextual information for capturing finer facial texture information. In addition, the carefully designed MSFE module and MLFF strategy to assist LGDF-Net for Deepfake detection. The MSFE can enhance the extraction of facial spatial features to minimize information loss during the network learning process. It provides a rich joint feature representation for the network by enhancing the synergy between local artifact features and global texture features at different levels. Experimental results demonstrate that the proposed LGDF-Net shows significant advantages in both detection accuracy and generalization compared to some state-of-the-art methods. Our future work will focus on inter-frame forge information

for deepfake detection based on the forging cues between consecutive frames in a Deepfake video.

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